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Question 1 [C03] [10 Points]

Given the following classes, design the **ExpressDelivery** class to generate the expected output.

Total Cost = Base Fee + (Weight × Rate per kg) + (Distance × Rate per km)

- For **Fast delivery**: Cost per kg 7 Tk and per km 3 Tk
- For **Saver delivery**: Cost per kg 5 Tk and per km 1 Tk

The estimated delivery time depends only on the distance: **2 days for less than 50 km, 3 days for between 50-70 km and 4 days for greater than 70 km.**

Driver Code	Expected Output
<pre>public class DeliveryTester { public static void main(String[] args) { ExpressDelivery e1 = new ExpressDelivery("Fast", "PKG101", 5, 23.5); ExpressDelivery e2 = new ExpressDelivery("Saver", "PKG102", 12, 55.5); ExpressDelivery e3 = new ExpressDelivery("Fast", "PKG103", 12, 84.5); ExpressDelivery [] ex = {e1, e2, e3}; for (int i = 0; i<ex.length; i++) { System.out.println(i+1+"-----"); ex[i].calculateCost(); ex[i].estimatedDays(); ex[i].packageInfo(); } } }</pre>	<pre>1----- Package ID: PKG101 Weight: 5.0 kg Distance: 23.5 km Type: Fast Cost: 155.5Tk Estimated Days: 2 2----- Package ID: PKG102 Weight: 12.0 kg Distance: 55.5 km Type: Saver Cost: 165.5Tk Estimated Days: 3 3----- Package ID: PKG103 Weight: 12.0 kg Distance: 84.5 km Type: Fast Cost: 387.5Tk Estimated Days: 4</pre>
<pre>public class Delivery { String packageId; double weight; double distance; int baseFee = 50; public Delivery(String packageId, double weight, double distance) { this.packageId = packageId; this.weight = weight; this.distance = distance; } public int baseFare(){ return baseFee; } public void printDetails() { System.out.println("Package ID: " + packageId); System.out.println("Weight: " + weight + " kg"); System.out.println("Distance: " + distance + " km"); } }</pre>	

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Question 1 [C03] [10 Points]

Given the following classes, design the **PremiumTicket** class to generate the expected output.

Total Cost = Base Charge + (Seats × Rate per seat) + (Duration × Rate per hour)

- For **VIP Ticket**: Cost per seat 500 Tk and per hour 100 Tk
- For **Regular Ticket**: Cost per seat 300 Tk and per hour 50 Tk

The estimated category depends on duration: **less than 3 hours -> Short event, between 3-5 hours Medium event and greater than 5 hours -> Long event.**

Driver Code	Expected Output
<pre> public class TicketTester { public static void main(String[] args) { PremiumTicket t1 = new PremiumTicket("VIP", "T101", 2, 2.5); PremiumTicket t2 = new PremiumTicket("Regular", "T102", 4, 4.0); PremiumTicket t3 = new PremiumTicket("VIP", "T103", 3, 6.0); PremiumTicket [] t = {t1, t2, t3}; for (int i = 0; i<t.length; i++) { System.out.println(i+1+"-----"); t[i].calculateCost(); t[i].estimatedCategory(); t[i].ticketInfo(); } } } </pre>	<pre> 1----- Ticket ID: T101 Seats: 2 Duration: 2.5 hours Type: VIP Cost: 1750.0Tk Category: Short Event 2----- Ticket ID: T102 Seats: 4 Duration: 4.0 hours Type: Regular Cost: 1900.0Tk Category: Medium Event 3----- Ticket ID: T103 Seats: 3 Duration: 6.0 hours Type: VIP Cost: 2600.0Tk Category: Long Event </pre>
<pre> public class Ticket { String ticketId; int seats; double duration; int baseCharge = 500; public Ticket(String ticketId, int seats, double duration) { this.ticketId = ticketId; this.seats = seats; this.duration = duration; } public int baseFare(){ return baseCharge; } public void printDetails() { System.out.println("Ticket ID: " + ticketId); System.out.println("Seats: " + seats); System.out.println("Duration: " + duration + " hours"); } } </pre>	